Notes of Model Identification and Data Analysis Modulo 2

A series of notes on the course MIDA Modulo 2 as taught in Politecnico di Milano by Savaresi during the academic year 2018/2019.

Notes of Model Identification and Data Analysis Modulo 2
- Childhood Memories
- Write the system of difference equations
- Compute the system transfer function
  - First Method - z Transformation
  - Second Method - Transformation Formula
- Compute the first 4 impulse response elements
  - First Method - From Difference Equations
  - Second Method - Matrix Multiplication Formula
  - Third Method - Long Division
  - Fourth Method - Geometric Series Trick
- Check the system observability and reachability
  - Observability
  - Reachability
- Compute the Hankel Matrix of order n
  - Definition
  - Shortcut
- Given the input response compute the transfer function
- Write the state space system in control form
- Apply the change of variable $\tilde{x} = Tx$ with $T = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ to the difference equations
- Identify the system matrices using the 4SID method

Doubts

Childhood Memories

- Geometric Series

$$\sum_{k=0}^{\infty} a^i = \frac{1}{1-a}$$

- Matrix's Inverse

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & c_{23} & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n1} & c_{n2} & \dots & c_{nn} \end{bmatrix}^T$$

Write the system of difference equations

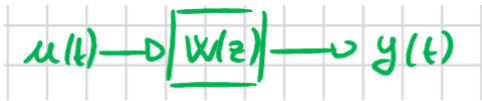
$$\begin{cases} x(t+1) = Fx(t) + Gu(t) \leftarrow \textit{state equation} \\ y(t) = Hx(t) + Ku(t) \qquad \leftarrow \textit{output equation} \end{cases}$$

$$x(t) = \begin{bmatrix} x_1(t) \\ . \\ . \\ . \\ x_n(t) \end{bmatrix} \in \mathbb{R}^n$$

$$y(t), u(t) \in \mathbb{R}^1$$

Compute the system transfer function

$$y(t) = W(z)u(t)$$



The transfer function is a representation of the input-output relationship.

There are two methods available.

First Method - z Transformation

Apply the z-transformation directly to the system of difference equations.

$$v(t+1) = z \cdot v(t)$$

Solve the system wrt to u.
Your goal is to end up with a representation of y in function of u.
The coefficient of u is W, the transfer function.

Second Method - Transformation Formula

We use the transformation formula.

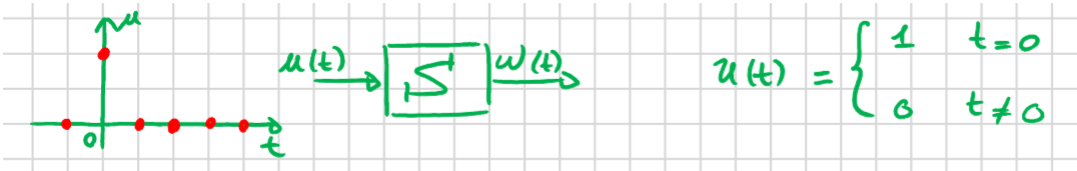
$$W(z) = H(zI - F)^{-1} + D$$

1. Compute $zI - F$
2. Compute $det(zI - F)$
3. Compute $(zI - F)^{-1}$
4. Compute $W(z) = H(zI - F)^{-1} + D$

Compute the first 4 impulse response elements

What is impulse response?

Simply the output you obtain after you have an impulse input in your system. the input is $u(t)$.



There are four methods to compute it.

First Method - From Difference Equations

From the system of difference equations.

Let's say for example the difference equations are

$$\begin{cases} x(t+1) = \frac{1}{2}x(t) + 2u(t) \\ y(t) = 3x(t) \end{cases}$$

The definition of impulse is the following:

$$u(t) = \begin{cases} 1 & t = 0 \\ 0 & t \neq 0 \end{cases}$$

if it is not specified consider the initial state equal to zero: $x(0) = 0$.

Let's do the math:

$$\begin{aligned}
t = 0 \qquad & x(0) = 0 & & y(0) = 3x(0) = 0 \\
t = 1 \qquad & x(1) = \frac{1}{2}x(0) + 2u(0) = 2 & & y(1) = 3x(1) = 6 \\
t = 2 \qquad & x(2) = \frac{1}{2}x(1) + 2u(1) = 1 & & y(2) = 3x(2) = 3 \\
t = 3 \qquad & x(3) = \frac{1}{2}x(2) + 2u(2) = \frac{1}{2} & & y(3) = 3x(3) = \frac{3}{2} \\
t = 4 \qquad & x(4) = \frac{1}{2}x(3) + 2u(3) = \frac{1}{4} & & y(4) = 3x(4) = \frac{3}{4} \\
t = 5 \qquad & x(5) = \frac{1}{2}x(4) + 2u(4) = \frac{1}{8} & & y(5) = 3x(5) = \frac{3}{8}
\end{aligned}$$

The impulse responses are given by the value of y at each timestep t:

$$
\begin{aligned}
w(0) &= 0; \\
w(1) &= 6; \\
w(2) &= 3; \\
w(3) &= \frac{3}{2}; \\
w(4) &= \frac{3}{4}; \\
w(5) &= \frac{3}{8};
\end{aligned}
$$

Second Method - Matrix Multiplication Formula

Matrix Multiplication formula.

Third Method - Long Division

Long division.

The output is represented like this:

$$y(t) = w(0)u(t) + w(1)u(t-1) + w(2)u(t-2) + \ldots$$

If we want the first 4 impulse responses we should write the above equation until the $w(4)u(t-4)$ component.

And we do the following:

$$y(t) = \left(w(0) + w(1)z^{-1} + w(2)z^{-2} + \ldots\right)u(t)$$

$$W(z) = \frac{B(z)}{A(z)} \underset{\substack{\uparrow \\ long\ division}}{=} w(0) + w(1)z^{-1} + w(2)z^{-2} + \ldots$$

We already know B and A.
Let's do the long division and equal it to the right member of the equation above.

Once done the long division you'll easily find the values for $w(0), w(1), w(2), w(3)$ and $w(4)$ just by looking at the coefficients of z terms.

You can stop doing the long division ass soon as you get the coefficients you are looking for, no need to go further obviously.

Fourth Method - Geometric Series Trick

Probably it's useful when the matrices are one-dimensional.
First of all we need to compute the transfer function.

$$W(z) = H(zI - F)^{-1} + D$$

You'll obtain a transfer function that hides a geometric series inside.
In an exercise I found:

$$W(z) = \cdots = \frac{6}{z - \frac{1}{2}} = \frac{6z^{-1}}{1 - \frac{1}{2}z^{-1}} = 6z^{-1}\left(\frac{1}{1 - \frac{1}{2}z^{-1}}\right) = 6z^{-1}\left(\sum_{k=0}^{\infty}\left(\frac{1}{2}z^{-1}\right)^k\right)$$

$$= 6z^{-1}\left(1 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2} + \frac{1}{8}z^{-3} + \frac{1}{16}z^{-4} + \frac{1}{32}z^{-5} + \dots\right) =$$

$$= 6z^{-1} + 3z^{-2} + \frac{3}{2}z^{-3} + \frac{3}{4}z^{-4} + \frac{3}{8}z^{-5} + \dots$$

$$\downarrow w(0) = 0; \quad w(1) = 6; \quad w(2) = 3; \quad w(3) = \frac{3}{2}; \quad w(4) = \frac{3}{4}; \quad w(5) = \frac{3}{8}$$

Check the system observability and reachability

Observability

J is observable if and only if the observability matrix θ is full rank $(det \neq 0)$.
Where

$$\theta = \begin{bmatrix} H \\ HF \\ HF^2 \\ \vdots \\ FF^{n-1} \end{bmatrix}$$

Reachability

J is reachable if and only if the reachability matrix $\mathscr{R}$ is full rank $(det \neq 0)$.
Where

$$\mathscr{R} = \begin{bmatrix} G & FG & F^2G & \dots & F^{n-1}G \end{bmatrix}$$

Compute the Hankel Matrix of order n

Definition

$$H_n = \begin{bmatrix} w(1) & w(2) & w(3) & \dots & w(n-1) & w(n) \\ w(2) & w(3) & w(4) & \dots & w(n) & w(n+1) \\ w(3) & w(4) & w(5) & \dots & w(n-2) & w(n-1) \\ \vdots & & & & & \\ w(n) & w(n+1) & w(n+2) & \dots & w(2n-2) & w(2n-1) \end{bmatrix}$$

Shortcut

$$H_n = \theta \mathscr{R}$$

Given the input response compute the transfer function

Let's explain it with an example:

Input Response:

$$w(t) = \begin{cases} 0 & t \leq 1 \\ (-2)^{2-t} & t > 1 \end{cases}$$

May be helpful to draw the input response from some instants t.

The thing we need to remember is that

$$y(t) = w(0) + w(1)u(t-1) + w(2)u(t-2) + \dots$$

so we know $w(t)$ $\forall t$ and we end up having

$$\begin{aligned} y(t) &= z^{-2}\left(1 - \frac{1}{2}z^{-1} + \frac{1}{4}z^{-4} + \dots\right)u(t) \\ &= z^{-2}\left(\sum_{k=0}^{\infty}\left(-\frac{1}{2}z^{-1}\right)^k\right)u(t) \\ &= z^{-2}\left(\frac{1}{1+\frac{1}{2}z^{-1}}\right)u(t) \end{aligned}$$

The blue colored part of the equation is our transfer function W.

Write the state space system in control form

The state space system in control form is given by

$$W(z) = \frac{b_0 z^{n-1} + b_1 z^{n-2} + \cdots + b_{n-1}}{z^n + a_1 z^{n-1} + \cdots + a_n}$$

(This means that we want the transfer function expressed with all exponents of the z terms greater or equal to zero).

$n =$the order of A.

$$F = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ -a_n & -a_{n-1} & \dots & \dots & \dots & \dots & \dots & -a_1 \end{bmatrix}$$

$$G = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

$$H = \begin{bmatrix} b_{n-1} & \dots & b_0 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 \end{bmatrix}$$

In order to check whether you have computed correctly such matrixes you can make sure that

$$W(z) = H(zI - F)^{-1} + D$$

Apply the change of variable $\tilde{x} = Tx$ with $T = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ to the difference equations

The difference equations are

$$\begin{cases} x(t+1) = Fx(t) + Gu(t) \\ y(t) = Hx(t) + Ku(t) \end{cases}$$

$$\tilde{x}(t) = Tx(t)$$

$$\begin{cases} \tilde{x}(t+1) = \tilde{F}\tilde{x}(t) + \tilde{G}u(t) \\ y(t) = \tilde{H}\tilde{x}(t) + \tilde{K}u(t) \end{cases}$$

$$x(t) = T^{-1}\tilde{x}(t)$$

$$\begin{cases} T^{-1}\tilde{x}(t+1) = FT^{-1}\tilde{x}(t) + Gu(t) \\ y(t) = HT^{-1}\tilde{x}(t) + Ku(t) \end{cases}$$

$$\downarrow$$

$$\begin{cases} \tilde{x}(t+1) = TFT^{-1}\tilde{x}(t) + TGu(t) \\ y(t) = HT^{-1}\tilde{x}(t) + Ku(t) \end{cases}$$

The first and the second systems (black one and blue one) are ***equivalent***: they have the same input output relationships.

All the transformations are done on the black system, the source one, in order to find the blue one, characterized by $\tilde{F}, \tilde{G}, \tilde{H}$ and $\tilde{K}$.

$$\begin{gathered} \tilde{F} = TFT^{-1} \\ \tilde{G} = TG \\ \tilde{H} = HT^{-1} \\ \tilde{K} = K \end{gathered}$$

Once we have such formulas we just need to compute the new matrices and get the new difference equations.

Identify the system matrices using the 4SID method

1. Identify the system order
2. Build H_{n+1}
3. Find a factorization of $H_{n+1} = \theta_{n+1}\mathscr{R}_{n+1}$
   How to do it?
   - put n independent rows of H_{n+1} in $\mathscr{R}_{n+1}$
   - fill the rows of θ_{n+1} such that $H_{n+1} = \theta_{n+1}\mathscr{R}_{n+1}$
4. Extract matrices $\hat{F}, \hat{G}, \hat{H}, \hat{D}$
   $\hat{F} = \theta_{n+1}(1:n,:)^{-1}\theta_{n+1}(2:n+1,:)$
   $\hat{G} = \mathscr{R}(:,1)$
   $\hat{H} = \theta_{n+1}(1,:)$
   $\hat{D} = 0$ (since the system is strictly proper)

We get difference matrices wrt the original ones $\rightarrow$ we have found an equivalent state space representation.

Doubts

- $$\begin{bmatrix} 0 & -\frac{1}{2} \end{bmatrix}$$

  $$\theta_3 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & -\frac{1}{2} \end{bmatrix}$$

  $$Q_3(:,1) = G = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

  $$Q_3(:,2) = FG = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{1}{2} \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1/2 \end{bmatrix}$$

  $$Q_3(:,3) = F^2G = F \cdot FG = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{1}{2} \end{bmatrix}\begin{bmatrix} 1 \\ -1/2 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/4 \end{bmatrix}$$

  $$Q_3 = \begin{bmatrix} 0 & 1 & -\frac{1}{2} \\ 1 & -\frac{1}{2} & 1/4 \end{bmatrix}$$

  $$\boxed{\theta_3 Q_3 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & -\frac{1}{2} \end{bmatrix}\begin{bmatrix} 0 & 1 & -\frac{1}{2} \\ 1 & -\frac{1}{2} & \frac{1}{4} \end{bmatrix} = \begin{bmatrix} 0 & 1 & -\frac{1}{2} \\ 1 & -\frac{1}{2} & \frac{1}{4} \\ -\frac{1}{2} & \frac{1}{4} & -\frac{1}{8} \end{bmatrix} = +I_3}$$

  (f) check that $F = \theta_3(1:m,:)^{-1}\,\theta_3(2:m+1,:)$

  $$\theta_3(1:m,:) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

  $$\theta_3(2:m+1,:) = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{1}{2} \end{bmatrix}$$

  non capisco la dicitura $\theta(1:n,:)$, non dovrebbe essere: considera le le riga dalla 1 alla n e tutte le colonne?